

Aaryan Kandiah

+65 87217899 | aaryan.kandiah@gmail.com | [aaryan126.github.io](https://github.com/aaryan126) | linkedin.com/in/aaryan-kandiah

Seeking part-time internships in Software and AI | **Availability:** May 2027 – August 2027

WORK EXPERIENCE

Univers, AI Innovation Engineer Intern

May 2025 – Apr

2026

- Engineered a **full-stack AI HVAC Agentic Platform** for internal GTM and engineering teams, featuring RAG architecture with VectorDB, context-aware recommendations, role-based AI reasoning, and AI-powered sales enablement tools **increasing GTM team productivity by ~25%**.
- Optimized **Java backend and React/TypeScript frontend** energy-saving dashboards for **8 clients**; debugged critical backend API endpoints, implemented real-time chatbot streaming, natural language data queries, and interactive visualizations, **reducing load time by 20%**.
- Deployed **2 on-premises AI HVAC solutions** on dedicated hardware with local LLM hosting, configuring end-to-end AI dashboards and backend/frontend services, enabling AI capabilities with zero cloud dependency.
- Built internal **AI monitoring service** to detect bugs and data anomalies across client-facing dashboards, with automated notifications to engineering team, **reducing incident response time by ~40%**.

EDUCATION

National University of Singapore (NUS) (Expected)

Aug 2026 – Dec 2027

- **Master of Computing (Artificial Intelligence)**

Nanyang Technological University (NTU)

Aug 2022 – Dec 2025

- **Bachelor of Engineering (Electrical and Electronic Engineering)**
- Specialization: **Data Analysis and Machine Learning**
- **Honours (Distinction)** CGPA: 4.36 / 5.00
- Relevant Modules: (1) *Data Structures and Algorithms* (2) *Image Processing and Computer Vision* (3) *Artificial Intelligence and Data Mining Intelligence* [**Deep Learning, Natural Language Programming**] (4) *Machine Learning Design & Application* [**Large Language Models, Transformer Architecture, Attention Mechanism**] (5) *Intelligent Systems Design* [**Deep Learning, Computer Vision, Statistical Pattern Recognition**]

Virtual Training and Learning & Development

- Amazon Web Services (AWS) Certification: *AI Practitioner*
- Coursera Relevant Module (Google): *Advanced Data Analytics Professional Certification*

ACADEMIC PROJECTS / COMPETITION PROJECT / RESEARCH EXPERIENCE

AI Workflow Competition, Echelon (Organized by e27) – **1st Place** (of 147 builders) Jun 2026

Project: **BOLDR SignalDesk – Approval-First AI Customer Intelligence Workflow for SME Support**

- Secured \$16,000 in prizes for architecting a human-in-the-loop AI platform automating SME customer support, spanning evidence-grounded reply drafting and marketing-intel generation from resolved queries.
- Developed **multi-source evidence retrieval pipeline** (FastAPI backend) unifying policy, product, and pricing documents into a single structured knowledge base for real-time reply drafting.
- Directed reply drafting and customer service resolutions via GLM-5.1-based orchestration layer with deterministic fallback logic, sustaining **~96% answerability accuracy**.
- Designed **self-improving knowledge loop** that converts human-resolved gaps into reusable knowledge-base entries, reducing repeat escalations and projecting **~24 hours/week** of support time saved at pilot scale.

Agent Forge AI Hackathon (Sponsored by Z.ai and GMI Cloud) – **1st Place** (of 147 builders) Apr 2026

Project: **Adapt – AI-Powered Ad Localization Platform for Southeast Asian Markets**

- Awarded **1st place** for building an end-to-end AI pipeline, localizing ads across **6 SEA countries**, generating culturally adapted strategies, multilingual copy, and localized imagery from text or image input.
- Architected a **multi-model orchestration system** (FastAPI + React) chaining GPT-5.4 (vision/text input), GLM-5.1 (strategy & copywriting), and Gemini 3.1 Flash (image generation) via GMI Cloud APIs.
- Implemented image-to-image editing via Gemini 3.1 Flash multi-turn contents format, enabling surgical edits (e.g., product swaps) that preserve original ad composition.
- Integrated Publer for **direct TikTok publishing**, enabling end-to-end workflow from ad ingestion to live social deployment in **under 3 minutes**.

NTU-EEE Module: **Final Year Project**

Jan 2025 – Dec 2025

Module Project: **Multi-Label Chest X-ray Abnormality Detection Using Deep Learning and Explainable AI**

- Developed **multi-label classification model** using modified DenseNet-121 in **PyTorch** to detect **14 thoracic abnormalities** (e.g., pneumonia, cardiomegaly) from the **NIH Chest X-ray dataset** (~112k images).
- Implemented **custom weighted loss function** to address class imbalance, **reducing false negatives by ~15%**.
- Enhanced performance through **data augmentation and batch normalization** for improved generalization.
- Achieved **average AUC score of 0.845**, matching performance of several state-of-the-art CXR studies.
- Integrated **Grad-CAM Explainable AI**, providing visualizations of X-ray regions influencing predictions, improving clinical interpretability.

SKILLS

- **Software Programming:** Python, JavaScript, TypeScript, SQL, Java

- **Frameworks/ Tools:** LangChain, PyTorch, Scikit-learn, Next.js, FastAPI, AWS, Docker, Kubernetes, Linux
- **Languages:** Proficient in English; conversant in Hindi and French